End-to-End Analysis of Sales Data

Data Visualization & Capstone Project

Data Science Internship, Week 4 | Prepared by Mayank Pal | 19 October 2026

118.73M	16.89M	14.23%	1,125,806
Total Sales	Total Profit	Profit Margin	Units Sold

Stakeholder summary

The business made 16.89M profit on 118.73M sales (a 14.23% margin) between Sep 2013 and Dec 2014. Nearly all of the profit comes from two segments, and one segment loses money. These are the points to take into the meeting:

- **Two segments carry the business.** Government (11.39M) and Small Business (4.14M) make 92% of all profit.
- **Enterprise loses money.** Its profit is -0.61M (below zero). Its costs (20.23M) are higher than its sales (19.61M). All 58 loss-making sales are Enterprise deals, and every one had a Medium or High discount.
- **Big discounts cost profit.** High-discount sales bring 37.37M in sales but keep only 9.1% as profit. Low-discount sales keep 17.9%.
- **Paseo is the best product.** 4.80M profit, about 28% of the total. All six products make a profit.
- **Countries are close in sales** (20.95M to 25.03M). The USA is first in sales but only fourth in profit. France is first in profit (3.78M).
- **Sales peak at the end of the year.** Oct 2014 (12.38M) and Dec 2014 (12.00M) were the best months. Sep to Dec 2014 sales were 37% above Sep to Dec 2013.

What we recommend

- Fix Enterprise first: review its prices, costs and discounts.
- Put a limit on High discounts.
- Protect Government and Small Business, and push Paseo.
- Get stock and staff ready for Oct to Dec.

1. Objective

Find out what drives profit in the sales data, where money is lost, and what the business should do about it. The results are shown in charts, a Power BI dashboard and this report.

2. Data and method

Item	Details
Data	700 sales rows, 16 columns (segment, country, product, discount band, units, prices, sales, COGS, profit, date)
Period	Sep 2013 to Dec 2014 (16 months). 2013 has only Sep to Dec.
Cleaning (Python)	Removed spaces in column names, filled the missing Discount Band with "None", converted Date to a real date, added a Profit Margin % column, saved cleaned_sales.csv
Analysis (Pandas)	Totals, group-by tables for segment, product, country and discount band, loss-making rows, Sep to Dec growth, pivot table of profit by segment and product
Charts	Python (Matplotlib/Seaborn), Excel (3 pivot tables with charts), Power BI
Model	Linear regression to predict Sales from Units Sold and Sale Price

3. Power BI dashboard

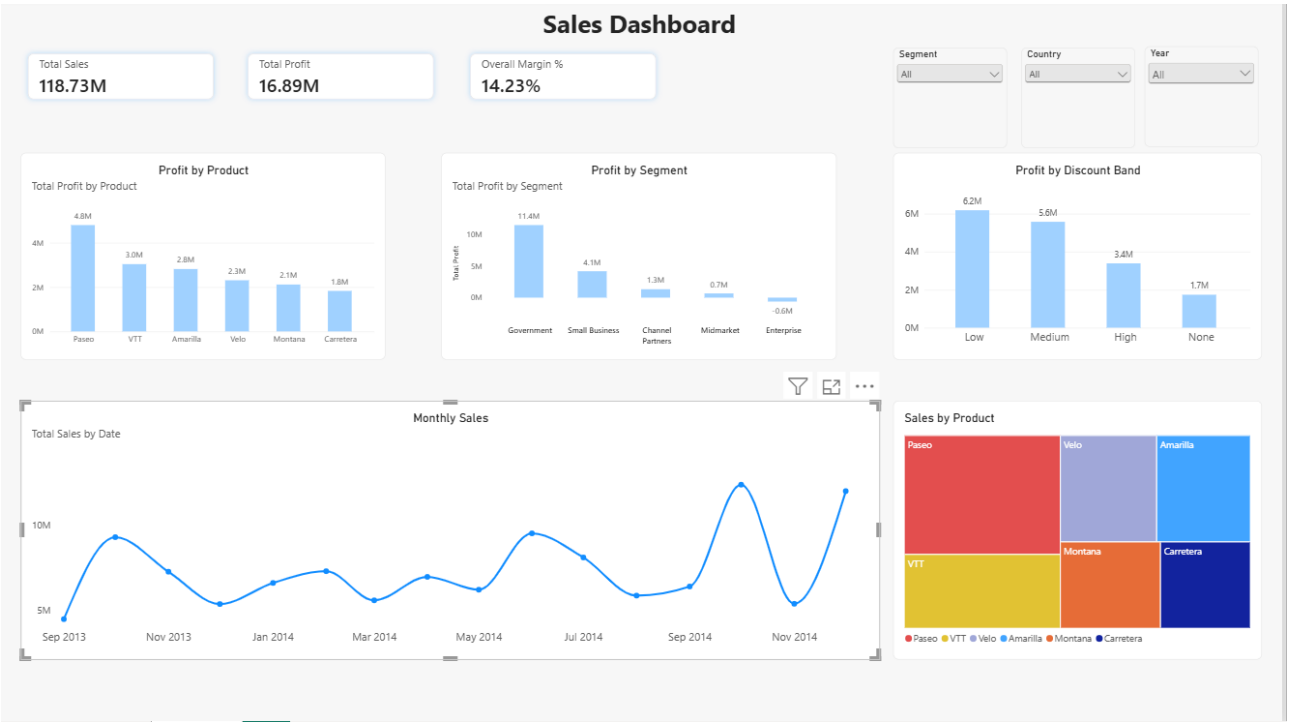


Figure 1. Sales dashboard. Three cards (sales, profit, margin), profit by product, segment and discount band, monthly sales, sales by product, and filters for segment, country and year.

The cards give the totals. The column charts show where profit comes from and where it is lost. The line chart shows the trend over time. The three drop-down filters let the reader look at one segment, country or year at a time.

4. Key findings

4.1 Profit by segment

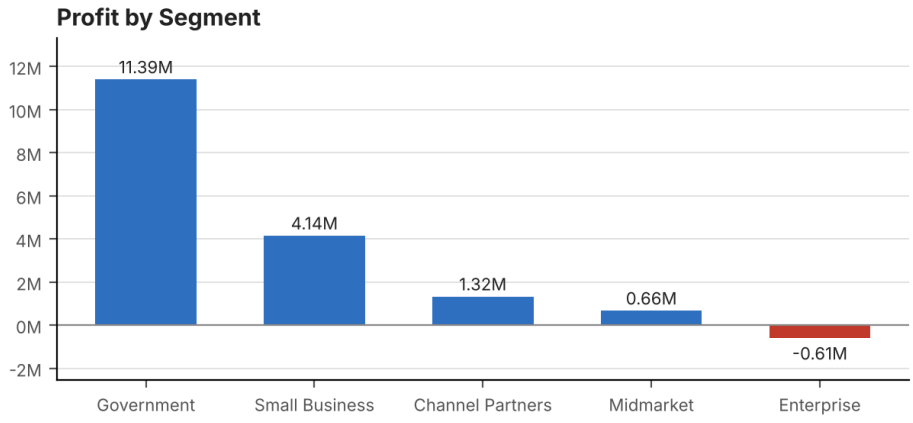


Figure 2. Profit by segment (Enterprise in red).

- Government earns 11.39M profit on 52.50M sales, a margin of 21.7% (total profit divided by total sales).
- Government and Small Business together make 92% of the profit.
- Enterprise loses 0.61M. Its margin is -3.1%, and its costs are 103% of its sales.
- All 58 loss-making rows are Enterprise: 33 with a High discount and 25 with a Medium discount.

4.2 Profit by product

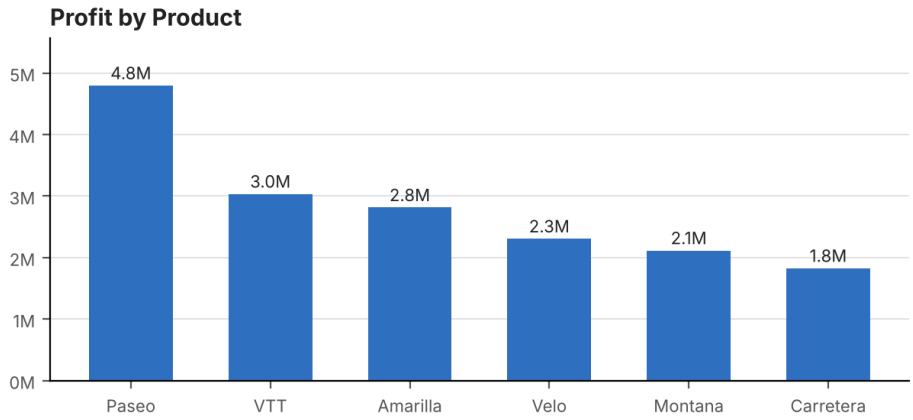


Figure 3. Profit by product.

- Paseo is first with 4.80M, then VTT (3.03M) and Amarilla (2.81M).
- Carretera is last with 1.83M, but still makes a profit.

4.3 Discounts

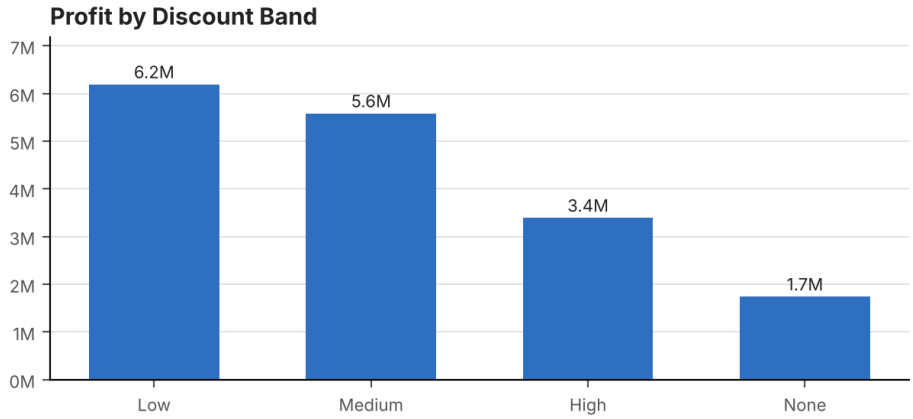


Figure 4. Profit by discount band.

Discount band	Low	Medium	High	None
Sales	34.63M	38.78M	37.37M	7.94M
Profit	6.19M	5.58M	3.39M	1.74M
Profit margin	17.9%	14.4%	9.1%	21.9%

- High-discount sales are as large as Low-discount sales (37.37M against 34.63M), but they give about half the profit (3.39M against 6.19M).
- The more the business discounts, the less of each sale it keeps.

4.4 Countries

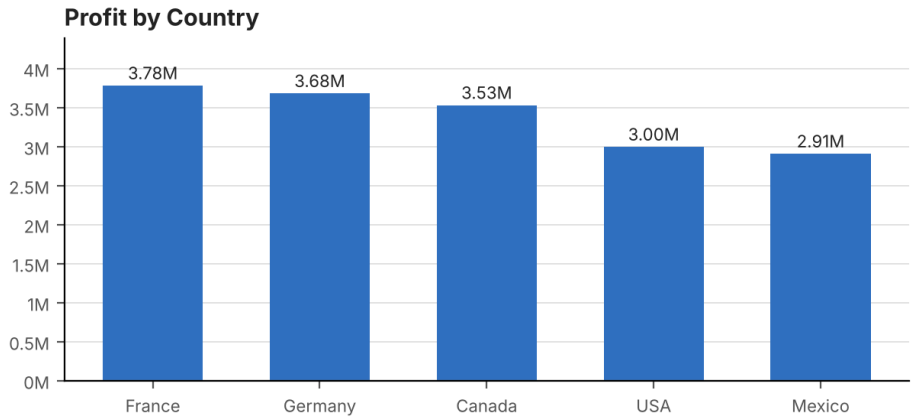


Figure 5. Profit by country.

- Sales are close: USA 25.03M, Canada 24.89M, France 24.35M, Germany 23.51M, Mexico 20.95M.
- Profit is not in the same order. France is first (3.78M), the USA is fourth (3.00M) and Mexico is last (2.91M).

4.5 Sales over time

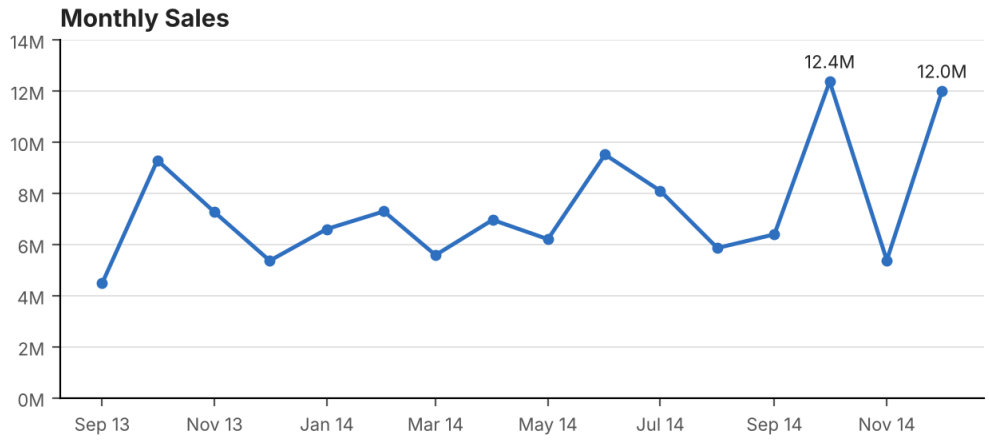


Figure 6. Monthly sales, Sep 2013 to Dec 2014.

- The best months are Oct 2014 (12.38M) and Dec 2014 (12.00M). Jun 2014 (9.52M) is the best month outside the end of the year.
- Sep to Dec 2014 sales were 36.16M, against 26.42M for Sep to Dec 2013 (+36.9%). Profit grew 40.1%.
- Sales are uneven. Nov 2014 fell to 5.38M between two very high months.

5. Prediction model

A simple linear regression predicts Sales from Units Sold and Sale Price. The data was split 80% for training (560 rows) and 20% for testing (140 rows).

Measure	Result	What it means
R-squared	0.762	The model explains about 76% of the changes in sales
Mean absolute error	83,636	On average the prediction is off by about this much per sale
Example	370,638	Predicted sales for 1,000 units at a sale price of 300

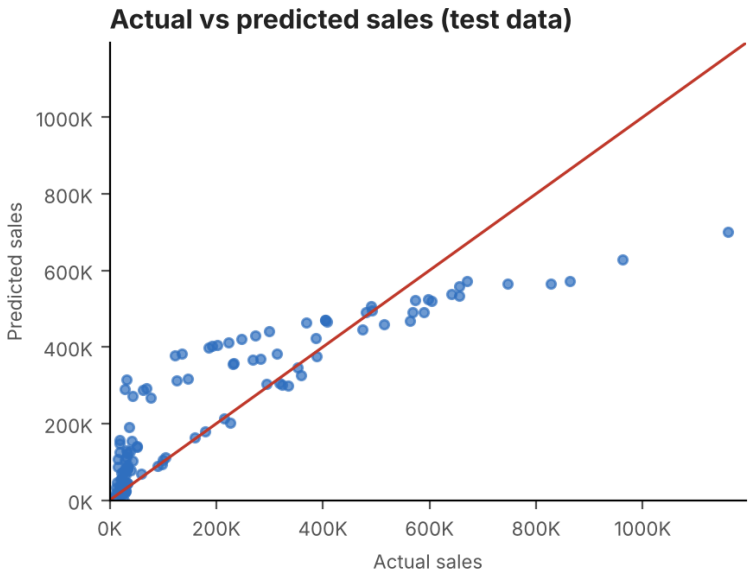


Figure 7. Actual against predicted sales on the test data. Points close to the red line are good predictions.

The fit is useful but not exact. Sales is really units multiplied by price, and a straight-line model adds the two instead. It is a starting point, not a forecast to plan on.

6. Recommendations

#	Recommendation	Because
1	Review Enterprise prices, costs and discounts	It loses 0.61M, and its costs are higher than its sales
2	Set a limit on High discounts	High discount keeps only 9.1% of sales as profit, against 17.9% for Low
3	Protect Government and Small Business, and push Paseo	Government and Small Business make 92% of profit, and Paseo is the top product
4	Look at why the USA profit is lower than its sales suggest	The USA is first in sales but fourth in profit
5	Plan stock and staff for Oct to Dec	The two biggest months are Oct and Dec 2014

7. Conclusion

The business is profitable (14.23% margin) but depends on two segments. The main problem is Enterprise, which loses money because its costs and discounts are too high. Fixing Enterprise pricing and limiting big discounts are the two changes most likely to raise profit.

8. Limits of this analysis

- The data has 700 rows and covers 16 months. 2013 has only four months, so only Sep to Dec can be compared across years.
- The results describe what happened. They do not prove why it happened.
- The regression is a simple model. The error is large for big sales.
- Segment margins in this report are total profit divided by total sales. The notebook (Question 6) averages the margin of each single sale, so its numbers differ (for example Government is 21.7% here and 29% there).

9. Files in the Week 4 folder

File	What it is
Data_Visualization_Capstone_Project.ipynb	Python notebook: cleaning, exploration, analysis, charts, regression, summary
cleaned_sales.csv	Cleaned dataset used for Excel and Power BI
Project_charts.xlsx	Excel file with 3 pivot tables and charts (segment, product, monthly sales)
Project_dashboard.pbix	Power BI dashboard
stakeholder_summary.pdf	One-page summary for the stakeholder presentation
Week4_project_report.pdf	This report